

**Proposed Data Centre
Development at No. 2-10,
Tai Yuen Street, Kwai
Chung, N.T.**

Technical Particular
Specifications for Monitoring

2020-10-19

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1. General

- 1.1 Prior to commencement of construction as directed by the Architect, the Contractor shall be responsible for installing additional instrumentation and, building and ground settlement points as shown on the drawings.
- 1.2 The Contractor shall satisfy himself that the drillhole locations for the instrumentation are free from existing services prior to commencement of drilling. In order to avoid any utilities and services, the required drillhole positions may be adjusted on site subject to the approval of the Architect. The Contractor shall be responsible for any damage, and any claims arising from such damage and shall make every effort to make good the damage as quickly as possible.
- 1.3 The Contractor shall satisfy himself that all existing instrumentation are functioning before he takes possession of site. He shall notify the Architect of any damaged instrumentation and replacement of instrumentation where required.
- 1.4 The Contractor shall be responsible for obtaining any necessary permits and permissions from the Relevant Authorities required in connection with the drilling work for the purpose of instrumentation installation, and excavation work for settlement point installation prior to commencement of the work.

2. Inclinometer System Installation

- 2.1 Inclinometers shall be installed in accordance with the written instructions of the Manufacturer of the equipment. Details of the equipment and manufacturer's instructions shall be submitted to the Architect for approval prior to commencement of installation.
- 2.2 Monitoring of inclinometers shall be carried out by an experienced and qualified personnel approved by the manufacturer of the equipment.

3. Piezometer and Standpipe Installation

- 3.1 Piezometer and standpipe shall be installed in accordance with the details as shown in the drawings or as agreed with the Architect.
- 3.2 Within two working days of the completion of the installation of the installation of any piezometer or standpipe, the Contractor shall carry out a response test to prove the successful operation of the installation. The test shall be carried out in the following manner:
 - (1) Record water level in piezometer/standpipe.
 - (2) Fill the piezometer/standpipe tube to the top with water.
 - (3) Record the fall of the water level at a distance from the top of the tube after the following elapsed time in minutes :
0, 1/4, 1/2, 1, 2, 4, 8, 15, 30, 45, 60 and thereafter at 30 minutes intervals until the water level has returned to its initial level.
 - (4) Results of all response tests shall be submitted to the Architect within 2 days of completion of each test for approval.

4. Ground Settlement and Building Settlement Point Installation

- 4.1 Ground settlement marker and building settlement marker shall be installed in accordance with the details as shown in the drawings or as agreed with the Architect.

5. Monitoring

- 5.1 Before commencement of excavation, the Contractor shall take initial readings of all monitoring points and submit initial readings to the Architect for record. Piezometer and standpipe readings shall be taken for at least 1 week to determine the average water table around the site to allow for fluctuation in water table.
- 5.2 The Contractor shall take monitoring readings at the frequency as specified by the Architect.

6. Records

- 6.1 The Contractor shall submit monitoring records to the Architect at weekly interval. All monitoring records shall be kept on site for inspection at all times.

7. Defective Instrumentation

- 7.1 All instrumentation shall be installed and functioning to the satisfaction of the Architect.
- 7.2 In the event the monitoring instrument is damaged due to whatever causes, the Contractor is responsible for repairing or replacing the damaged instrumentation to the satisfaction of the Architect.

8. Vibration Monitoring

- 8.1 Where required, the Contractor shall provide a vibrograph and any other necessary instrument to measure the effects of vibration generated by his works.
- 8.2 When excessive movement, water table drawdown or vibration is observed, works shall cease. The Contractor shall submit his proposal for remedial measures to rectify the situation before recommencement of works.
- 8.3 Vibration caused by piling work shall not exceed the maximum peak particle velocity of ground movement of 7.5 mm/second and 15 mm/second for water retaining structures and water mains respectively, as measured with a vibroscope. Comply with restrictions on particle velocity caused to adjoining structures as specified.

9. Tiltmeter System

- 8.4 Tilt-plates and tiltmeters shall be installed in accordance with the written instructions of the Manufacturer of the equipment. Details of the equipment and manufacturer's instructions shall be submitted to the Architect for approval prior to commencement of installation.
- 8.5 Monitoring of tiltmeters shall be carried out by an experienced and qualified personnel approved by the manufacturer of the equipment.

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